

Mathematics Stage 3 -- Year B -- Unit 21

Number: Place Value, Decimals & Powers of 10

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly.
MA3-RN-01	Applies an understanding of place value and the role of zero to represent the properties of numbers.
MA3-RN-02	Compares and orders decimals up to 3 decimal places.
MA3-AR-01	Selects and applies appropriate strategies to solve addition and subtraction problems.
MA3-MR-01	Selects and applies appropriate strategies to solve multiplication and division problems.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- Applying	Green -- Extension
Place value & decimals (RN-A) + ×10/100/1000 (MR-A) Arrange numbers in the millions in ascending and descending order using place value; recognise 1000 thousands is 1 million and 1000 millions is 1 billion (RN-A). Compare and order decimals to 3 dp; interpret trailing zeros; approximate the size of decimals; place decimals on a number line (RN-A). Use mental strategies to multiply one-digit numbers by 10, 100 and 1000 and their multiples (MR-A).	Partitioning, place-value shifts & decimal operations (applying) Regroup numbers in different forms and partition numbers to 1 billion in non-standard forms (RN-A). Determine numbers that are 10 or 100 times a decimal, and one-tenth or one-hundredth of it, using place value (RN-A). Model the addition and subtraction of decimals up to 3 decimal places (AR-B). Estimate the product of 2 numbers using multiples of 10 or 100, and use estimation to check the reasonableness of answers (MR-A).	Extended -- Stage 4 Indices, Decimals & Order of Operations Represent numbers in index notation (positive powers) and evaluate powers of 10 (MA4-IND-C-01). Multiply and divide decimals by powers of 10; round decimals to a specified number of decimal places (MA4-FRC-C-01). Apply the order of operations to evaluate expressions; use estimation and rounding to check answers (MA4-INT-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students read, write and say large numbers and decimals, and explain their place-value reasoning and chosen strategies.
Understanding and fluency	Students order numbers and decimals, multiply by 10/100/1000, partition numbers, shift decimals by powers of 10 and add/subtract decimals.
Reasoning	Students reason about the value of trailing zeros, why a place-value shift changes a digit's value, and whether an estimate is reasonable.
Problem solving	Students partition in non-standard forms, estimate products to check answers, and (Stage 4) work with index notation, decimal multiplication/division and the order of operations.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key finding s	

Post-test / unit evaluation

Date	
Key finding s	
Stu- den- ts who need further consol- idation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 21 | NESA Mathematics K-10 Syllabus (2022)

Unit 21 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Read and order numbers in the millions.	RN-A: arrange numbers in the millions in ascending/descending order; 1000 thousands = 1 million, 1000 millions = 1 billion MA3-RN-01 Year A	Mini Lesson 1		
R2	Compare and order decimals to 3 decimal places.	RN-A: compare/order decimals to 3 dp; interpret trailing zeros; approximate decimal size; place on a number line MA3-RN-02 Year A	Mini Lesson 2		
R3	Multiply by 10, 100 and 1000 using mental strategies.	MR-A: use mental strategies to multiply one-digit numbers by 10, 100, 1000 and their multiples MA3-MR-01 Year A	Mini Lesson 3		
Y1	Partition and regroup numbers in non-standard forms.	RN-A: regroup numbers in different forms; partition numbers to 1 billion in non-standard forms MA3-RN-01 Year A	Mini Lesson 4		
Y2	Find 10/100 times (and one-tenth/one-hundredth of) a decimal.	RN-A: determine numbers that are 10 or 100 times the original decimal, and one-tenth or one-hundredth of it MA3-RN-02 Year A	Mini Lesson 5		
Y3	Add and subtract decimals to 3 decimal places.	AR-B: model the addition and subtraction of decimals up to 3 dp using appropriate representations MA3-AR-01 Year B	Mini Lesson 6		
Y4	Estimate products and check the reasonableness of answers.	MR-A: estimate the product of 2 numbers using multiples of 10 or 100; use estimation to check reasonableness MA3-MR-01 Year A	Mini Lesson 7		
G1	Powers of 10 and index notation.	Stage 4 IND: represent numbers in index notation (positive powers); evaluate numbers in index notation, including powers of 10 MA4-IND-C-01 Stage 4 (Core)	Mini Lesson 8		
G2	Multiply and divide decimals.	Stage 4 FRC: multiply and divide decimals by powers of 10; round decimals to a specified number of places MA4-FRC-C-01 Stage 4 (Core)	Mini Lesson 9		
G3	Order of operations with whole numbers and decimals.	Stage 4 INT: apply the order of operations to evaluate expressions; estimate/round to check answers MA4-INT-C-01 Stage 4 (Core)	Mini Lesson 10		

Syllabus outcomes and content from Mathematics K-10 Syllabus (2022) copyright NSW Education Standards Authority (NESA) for and on behalf of the Crown in right of the State of New South Wales, 2024. RN = Represents numbers. AR = Additive relations. MR = Multiplicative relations. A/B = syllabus content group (Year A ~ Year 5; Year B ~ Year 6).

Unit 21 -- Mini Lesson Sequence

Number: Place Value, Decimals & Powers of 10 | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-3

Yellow band -- On-level -- Mini Lessons 4-7

Green band -- Extension -- Mini Lessons 8-10

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 21. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Reading and ordering numbers in the millions

R1 | Red band | DoE Lesson 1 (adapted)

Duration: 25-30 min **Outcomes:** R1 Syllabus: MA3-RN-01, MAO-WM-01 | RN-A: order numbers in the millions

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Applying place value (numbers in the millions) | Key resources: Place value charts (to hundred-millions), digit cards

Learning intention

Students are learning to:
read and order numbers in the millions using place value

Success criteria

Students can:

- * I can read a number in the millions using a place value chart.
- * I can order numbers in the millions in ascending and descending order.
- * I know 1000 thousands is 1 million and 1000 millions is 1 billion.

Mini masterclass structure (25-30 min)

Activate (4 min): Write 3 450 000 on the board. How do we read it? Which digit is in the millions place?

Model (10 min): Use a place value chart with periods (millions, thousands, ones). Compare numbers by the highest place value first. Build up: 1000 thousands = 1 million; 1000 millions = 1 billion.

Guided practice (10 min): Students order sets of numbers in the millions (ascending and descending) using place value.

Check (4 min): Exit question: order 1 200 000, 1 020 000 and 1 002 000 from smallest to largest.

Key vocabulary

place value, millions, billion, period, digit, ascending, descending, order

Differentiation

Support: Provide a place value chart with labelled columns so digits line up.
Extension: Order numbers up to 1 billion, including numbers with internal zeros.

Assessment

Students read and order numbers in the millions — links to R1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Comparing and ordering decimals [HANDS-ON]

R2 | Red band | DoE Lesson 4 (adapted)

Duration: 25-30 min **Outcomes:** R2 Syllabus: MA3-RN-02, MAO-WM-01 | RN-A: compare/order decimals to 3 dp

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Place value and positioning of decimals | Key resources: Decimal number lines, place value charts

Physical materials needed: decimal number lines, place value charts

Learning intention		Success criteria
Students are learning to: compare and order decimals to 3 decimal places and place them on a number line		Students can: * I can compare decimals digit by digit using place value. * I know a zero at the end of a decimal (e.g. 0.40) does not change its value. * I can place decimals to 3 decimal places on a number line and approximate their size.
Mini masterclass structure (25-30 min)		
Activate (4 min): Which is bigger, 0.4 or 0.39? Ask students to justify using place value. Model (10 min): Line up the decimal points; compare tenths, then hundredths, then thousandths. Show $0.4 = 0.40 = 0.400$. Place several decimals on a number line and approximate (0.62 is close to 0.6). Guided practice (10 min): Students order sets of decimals to 3 dp and place them on a number line. Check (4 min): Exit question: order 0.7, 0.07, 0.77 and 0.707 from smallest to largest.		
Key vocabulary		Differentiation
decimal, place value, tenths, hundredths, thousandths, compare, order, number line, approximate		Support: Use a place value chart so digits line up under the right column. Extension: Name a decimal that lies between two given decimals (e.g. between 0.4 and 0.5).
Assessment		Teaching notes (annotate here)
Students compare and order decimals — links to R2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Multiplying by 10, 100 and 1000

R3 | Red band | DoE Lesson 5 (adapted)

Duration: 25 min | **Outcomes: R3** | Syllabus: MA3-MR-01, MAO-WM-01 | MR-A: multiply by 10, 100, 1000 mentally

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Multiplying and dividing whole numbers and decimals by 10, 100 and 1000 | Key resources: Place value charts

Learning intention		Success criteria	
Students are learning to: use mental strategies to multiply one-digit numbers by 10, 100 and 1000		Students can: * I know multiplying by 10 shifts each digit one place to the left. * I can multiply one-digit numbers by 10, 100 and 1000. * I can multiply by multiples like 30, 200 or 4000.	
Mini masterclass structure (25 min)			
Activate (3 min): What is 7×10 ? Ask students to explain what happens to the 7 on a place value chart. Model (10 min): On a place value chart, $\times 10$ shifts a digit one place left ($6 \rightarrow 60$), $\times 100$ two places ($6 \rightarrow 600$), $\times 1000$ three places ($6 \rightarrow 6000$). Extend to multiples: $4 \times 30 = 120$; $3 \times 200 = 600$. Guided practice (8 min): Students multiply one-digit numbers by 10, 100, 1000 and their multiples. Check (4 min): Exit question: $8 \times 1000 = ?$ and $7 \times 200 = ?$			
Key vocabulary		Differentiation	
place value, multiply, powers of 10, multiple, shift, mental strategy		Support: Use a place value chart and physically move the digit left. Extension: Multiply two-digit numbers by 10, 100 and 1000 (e.g. 24×100).	
Assessment		Teaching notes (annotate here)	
Students multiply by 10, 100 and 1000 — links to R3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 4 Partitioning and regrouping in non-standard forms

Y1 | Yellow band | DoE Lesson 2 (adapted)

Duration: 30 min | **Outcomes:** Y1 | Syllabus: MA3-RN-01, MAO-WM-01 | RN-A: regroup; partition in non-standard forms

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Partitioning numbers in non-standard forms | Key resources: Place value charts, expander cards

Learning intention		Success criteria	
Students are learning to: partition and regroup numbers in standard and non-standard forms		Students can: * I can partition a number in standard form (e.g. $4500 = 4000 + 500$). * I can partition a number in non-standard forms that show the same value. * I can regroup a number in different ways without changing its value.	
Mini masterclass structure (30 min)			
Activate (4 min): Partition 2300 in the usual way ($2000 + 300$). Is there another way to make 2300?			
Model (12 min): Standard: $4500 = 4000 + 500$. Non-standard: $4500 = 3000 + 1500 = 45 \text{ hundreds} = 4000 + 400 + 100$. Show each is still 4500 by recombining.			
Guided practice (10 min): Students write numbers (to 1 billion) in two or more non-standard forms and check they regroup to the same value.			
Check (4 min): Exit question: write 5200 in two different non-standard forms.			
Key vocabulary		Differentiation	
partition, regroup, standard form, non-standard form, place value, expanded form		Support: Use place value expander cards so parts physically recombine into the whole. Extension: Partition a 7-digit number (millions) in non-standard form.	
Assessment		Teaching notes (annotate here)	
Students partition and regroup numbers — links to Y1 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 5 Ten and one hundred times a decimal

Y2 | Yellow band | DoE Lesson 5 (adapted)

Duration: 25-30 min

Outcomes: Y2

Syllabus: MA3-RN-02, MAO-WM-01 | RN-A: 10/100 times a decimal; one-tenth/one-hundredth

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Multiplying and dividing whole numbers and decimals by 10, 100 and 1000 | Key resources: Place value charts

Learning intention

Students are learning to:
find the number that is 10 or 100 times a decimal, and one-tenth or one-hundredth of it

Success criteria

Students can:

- * I know 10 times a decimal shifts each digit one place to the left.
- * I know one-tenth of a decimal shifts each digit one place to the right.
- * I can find 100 times and one-hundredth of a decimal.

Mini masterclass structure (25-30 min)

Activate (3 min): What is 10 times 0.6? Ask students to predict and explain.

Model (12 min): On a place value chart: $0.06 \rightarrow 0.6 \rightarrow 6 \rightarrow 60$ (each $\times 10$ shifts left). One-tenth of $0.6 = 0.06$ (shift right). $100 \text{ times } 0.35 = 35$; one-hundredth of $0.35 = 0.0035$.

Guided practice (10 min): Students find 10 and 100 times a decimal, and one-tenth/one-hundredth of it, using place value.

Check (4 min): Exit question: $10 \text{ times } 0.35 = ?$ and one-tenth of $0.35 = ?$

Key vocabulary

place value, decimal, ten times, one hundred times, one-tenth, one-hundredth, shift

Differentiation

Support: Use a place value chart and slide a digit card left or right.

Extension: Chain shifts: find 1000 times a decimal, then one-thousandth of the result.

Assessment

Students apply place-value shifts to decimals — links to Y2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Adding and subtracting decimals [HANDS-ON]

Y3 | Yellow band | DoE Lesson 6 (adapted)

Duration: 30 min | **Outcomes:** Y3 | Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: model add/subtract decimals to 3 dp

Linked to: DoE Lesson 6 (adapted) -- Lesson 6 -- Operations with decimals | Key resources: Place value charts, number lines

Physical materials needed: place value charts, number lines

Learning intention		Success criteria	
Students are learning to: add and subtract decimals up to 3 decimal places, lining up the decimal points		Students can: * I can line up the decimal points to add or subtract decimals. * I can use zeros as place holders so both numbers have the same number of decimal places. * I can add and subtract decimals to 3 dp to solve problems.	
Mini masterclass structure (30 min)			
Activate (4 min): Buy items for \$3.50 and \$2.75. How much altogether? How would you set it out?			
Model (12 min): Line up the decimal points and add place by place (\$6.25). Pad with zeros: $3.25 + 1.4 = 3.250 + 1.400 = 4.650$. Subtract: $7.5 - 3.25 = 4.25$, modelled on a place value chart.			
Guided practice (10 min): Students add and subtract decimals to 3 dp, including money problems, lining up the points.			
Check (4 min): Exit question: $7.5 - 3.25 = ?$ Show how you lined it up.			
Key vocabulary		Differentiation	
decimal, add, subtract, line up, decimal point, place holder, place value		Support: Use a place value chart so the decimal points and columns line up. Extension: Solve a two-step money problem mixing addition and subtraction of decimals.	
Assessment		Teaching notes (annotate here)	
Students add and subtract decimals — links to Y3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 7 Estimating products and checking reasonableness

Y4 | Yellow band | DoE Lesson 8 (adapted)

Duration: 25-30 min **Outcomes:** Y4 Syllabus: MA3-MR-01, MAO-WM-01 | MR-A: estimate products; check reasonableness

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- Using a range of strategies to solve problems | Key resources: Rounding number lines

Learning intention		Success criteria
Students are learning to: estimate the product of two numbers and use estimation to check answers		Students can: * I can round a 2- or 3-digit number to the nearest 10 or 100. * I can estimate a product using multiples of 10 or 100. * I can use an estimate to check whether an answer is reasonable.
Mini masterclass structure (25-30 min)		
Activate (3 min): Roughly how much is 6×49? Ask for a quick estimate before any working. Model (12 min): Round 49 $\rightarrow$ 50, so $6 \times 49 \approx 6 \times 50 = 300$ (exact 294 — close, so reasonable). Round 280 $\rightarrow$ 300, so $3 \times 280 \approx 3 \times 300 = 900$. Compare estimate to exact. Guided practice (10 min): Students estimate products by rounding, then check given answers for reasonableness. Check (4 min): Exit question: estimate 7×62. Is $7 \times 62 = 434$ reasonable?		
Key vocabulary		Differentiation
estimate, product, round, multiple, reasonable, approximate, check		Support: Provide the rounded numbers so students focus on the estimate. Extension: Estimate a 2-digit by 2-digit product and explain how close the estimate is.
Assessment		Teaching notes (annotate here)
Students estimate products and check reasonableness — links to Y4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Powers of 10 and index notation

G1 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G1 | Syllabus: MA4-IND-C-01, MAO-WM-01 | Stage 4: index notation; powers of 10

Learning intention		Success criteria
Students are learning to: write powers of 10 in index notation and use them to write large numbers		Students can: * I can write a power of 10 in index notation (e.g. $10 \times 10 = 10^2$). * I know the index tells me how many tens are multiplied (and how many zeros 10 to that power has). * I can write and evaluate powers of 10 up to $10^6 = 1\,000\,000$.
Mini masterclass structure (30 min)		
Activate (4 min): $10 \times 10 = 100$. Is there a shorter way to write 'ten multiplied by itself'? Model (12 min): Introduce the base and index: $10^2 = 100$, $10^3 = 1000$, $10^6 = 1\,000\,000$. The index = the number of tens multiplied = the number of zeros. Evaluate $10^4 = 10\,000$. Guided practice (10 min): Students write powers of 10 in index notation and evaluate them in standard form. Check (4 min): Exit question: write 10^4 in standard form, and write 1 000 000 as a power of 10.		
Key vocabulary		Differentiation
index notation, power, exponent, base, powers of 10, standard form, evaluate		Support: Build a table linking the index, the zeros and the standard-form number. Extension: Write a large number such as 4 000 000 as 4×10^6.
Assessment		Teaching notes (annotate here)
Students use index notation for powers of 10 — extends R1 and R3.		Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Multiplying and dividing decimals

G2 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G2 | Syllabus: MA4-FRC-C-01, MAO-WM-01 | Stage 4: $\times/\div$ decimals by powers of 10; round to a given dp

Learning intention		Success criteria	
Students are learning to: multiply and divide decimals by powers of 10 and round to a given number of places		Students can: * I can multiply a decimal by 10, 100 and 1000 (shift digits left). * I can divide a decimal by 10, 100 and 1000 (shift digits right). * I can round a decimal to 1 or 2 decimal places.	
Mini masterclass structure (30 min)			
Activate (3 min): What is 3.45×100 ? Ask students to predict the digit shift. Model (12 min): $\times 100$ shifts digits two places left ($3.45 \rightarrow 345$). $\div 10$ shifts one place right ($3.45 \rightarrow 0.345$). Round 3.456 to 2 dp $\rightarrow 3.46$ (look at the third place). Guided practice (10 min): Students multiply and divide decimals by powers of 10 and round answers to a given number of places. Check (4 min): Exit question: $0.7 \times 1000 = ?$ and round 2.358 to 1 decimal place.			
Key vocabulary		Differentiation	
decimal, multiply, divide, powers of 10, round, decimal place, shift		Support: Use a place value chart and move every digit the same number of places. Extension: Divide a decimal by 1000 and round the result to 2 dp.	
Assessment		Teaching notes (annotate here)	
Students multiply/divide decimals and round — extends R2, R3 and Y3.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 10 Order of operations with whole numbers and decimals G3 Green band Original mini lesson (Stage 4 extension)		
Duration: 30 min	Outcomes: G3	Syllabus: MA4-INT-C-01, MAO-WM-01 Stage 4: order of operations; check with estimation
Learning intention		Success criteria
Students are learning to: use the order of operations to evaluate expressions		Students can: * I know we agree on an order so everyone gets the same answer. * I can do brackets first, then multiplication and division, then addition and subtraction. * I can use estimation to check that my answer is reasonable.
Mini masterclass structure (30 min)		
Activate (4 min): Work out $2 + 3 \times 4$ two different ways. Why do we need an agreed order? Model (12 min): Order of operations: brackets, then $\times$ and $\div$, then $+$ and $-$. So $2 + 3 \times 4 = 14$, but $(2 + 3) \times 4 = 20$ — brackets change the answer. Estimate to check a longer expression is reasonable. Guided practice (10 min): Students evaluate expressions with and without brackets, including simple decimals. Check (4 min): Exit question: work out $10 - (2 + 3) \times 1$ and $20 \div (5 - 1)$.		
Key vocabulary		Differentiation
order of operations, brackets, grouping symbols, evaluate, expression, estimate		Support: Provide the operation order as a checklist to follow each time. Extension: Place brackets in a number sentence to make it equal a given target.
Assessment		Teaching notes (annotate here)
Students apply the order of operations — extends Y4.		Taught as described / adapted / replaced -- note here:

Unit 21 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Read and order numbers in the millions.		
Video	Place Value to Millions Mrs Roberts Resources	https://www.youtube.com/watch?v=yzw3j-O9AYY
Video	Reading and Writing Numbers to the Hundred Millions Easy Math with Mrs. Easley	https://www.youtube.com/watch?v=nMvHx_ctugw
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
R2 -- Compare and order decimals to 3 decimal places.		
Video	Comparing and Ordering Decimals Math with Mr. J	https://www.youtube.com/watch?v=KWsLyjXKhrM
Video	How to Order Decimals Math with Mr. J	https://www.youtube.com/watch?v=1dfedgVxA54
Worksheet (print)	Corbettmaths — Ordering Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/ordering-decimals-pdf.pdf
R3 -- Multiply by 10, 100 and 1000 using mental strategies.		
Video	Multiplying Multiples of 10 — Fast Mental Math Math with Mr. J	https://www.youtube.com/watch?v=ly7q52L3yN0
Video	Multiplying and Dividing by Powers of 10 Math with Mr. J	https://www.youtube.com/watch?v=HH-yrNS80Cg
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
Y1 -- Partition and regroup numbers in non-standard forms.		
Video	Whole Number Expanded Form, Word Form and Standard Form Math with Mr. J	https://www.youtube.com/watch?v=NOx6jIVSSpU
Video	Writing Numbers in Expanded Form (Millions) Elementary Math with Mr. J	https://www.youtube.com/watch?v=Ok_zzg75hO8
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
Y2 -- Find 10/100 times (and one-tenth/one-hundredth of) a decimal.		
Video	Multiplying Decimals by 10, 100, 1000 Lets Do Math	https://www.youtube.com/watch?v=GfL0s6tCNV4
Video	Dividing by Powers of Ten Math with Mr. J	https://www.youtube.com/watch?v=E8b5pDvt2AE
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
Y3 -- Add and subtract decimals to 3 decimal places.		
Video	Adding and Subtracting Decimals (How to) Math with Mr. J	https://www.youtube.com/watch?v=PnwLv6khwk8
Video	How to Add and Subtract Decimals (Step-by-Step Examples) Math with Mr. J	https://www.youtube.com/watch?v=Cg-TeiaSa8
Worksheet (print)	Corbettmaths — Adding Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/adding-decimals-pdf.pdf
Worksheet (print)	Corbettmaths — Subtracting Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/subtracting-decimals-pdf.pdf
Y4 -- Estimate products and check the reasonableness of answers.		
Video	Estimating Whole Number Products Multiplication Estimation Math with Mr. J	https://www.youtube.com/watch?v=mBWr8c0Lsx4
Video	Estimating Whole Number Sums and Differences Math with Mr. J	https://www.youtube.com/watch?v=EvQf38lnAJc
Worksheet (print)	Corbettmaths — Estimation (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/estimation-pdf.pdf
G1 -- Powers of 10 and index notation.		
Video	What are Powers of 10? Exponential, Expanded and Standard Form Math with Mr. J	https://www.youtube.com/watch?v=SKruiXYeFcI
Video	What is an Exponent? An Intro to Exponents Math with Mr. J	https://www.youtube.com/watch?v=NS4vHqJIPiE
Worksheet (print)	Corbettmaths — Indices (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/indices-pdf.pdf

Type	Resource	Link
G2 -- Multiply and divide decimals.		
Video	Multiply a Whole Number by a Decimal Math with Mr. J	https://www.youtube.com/watch?v=tsOibhsgYoQ
Video	Multiplying and Dividing by Powers of 10 Math with Mr. J	https://www.youtube.com/watch?v=HH-yrNS80Cg
Worksheet (print)	Corbettmaths — Multiplying and Dividing Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/decimals-multiplying-and-dividing.pdf
G3 -- Order of operations with whole numbers and decimals.		
Video	Order of Operations — A Step-By-Step Guide PEMDAS Math with Mr. J	https://www.youtube.com/watch?v=x41BCrYh8Kc
Video	Order of Operations — Parentheses, Brackets and Braces PEMDAS Math with Mr. J	https://www.youtube.com/watch?v=u7Lack4Tmx8
Worksheet (print)	Corbettmaths — Order of Operations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	What are Powers of 10? Exponential, Expanded and Standard Form Math with Mr. J	https://www.youtube.com/watch?v=SKruiXYeFcl
Video	Multiply a Whole Number by a Decimal Math with Mr. J	https://www.youtube.com/watch?v=tsOibhsgYoQ
Video	Order of Operations — A Step-By-Step Guide PEMDAS Math with Mr. J	https://www.youtube.com/watch?v=x41BCrYh8Kc
Worksheet (print)	Corbettmaths — Indices (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/indices-pdf.pdf
Worksheet (print)	Corbettmaths — Multiplying and Dividing Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/decimals-multiplying-and-dividing.pdf

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Place Value (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf>
- ☐ Corbettmaths — Ordering Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/ordering-decimals-pdf.pdf>
- ☐ Corbettmaths — Adding Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/adding-decimals-pdf.pdf>
- ☐ Corbettmaths — Subtracting Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/subtracting-decimals-pdf.pdf>
- ☐ Corbettmaths — Estimation (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/estimation-pdf.pdf>
- ☐ Corbettmaths — Indices (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/indices-pdf.pdf>
- ☐ Corbettmaths — Multiplying and Dividing Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/decimals-multiplying-and-dividing.pdf>
- ☐ Corbettmaths — Order of Operations (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf>